

Homework 24

This is our last homework assignment of the semester! Thanks for sticking with me to the end.

Please read Section 6.6 of Durrett. Please also look at Theorem 6.7, which I mentioned in class.

Please work the following exercises from Chapter 6.

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Let X be a random variable representing the total yearly claim for the 10,000 policy holders. We have $X = X_1 + \cdots + X_{10000}$, where the random variable X_i represents the yearly claim for policy holder i . Durrett wants us to make the (bad!) assumption that the random variables X_i are independent. With this assumption, since X is the sum of a large number of iid random variables, it is reasonable to expect that X is approximately normally distributed.

We have $E[X_1 + \cdots + X_{10000}] = E[X_1] + \cdots + E[X_{10000}]$ by additivity of expected value. Since the X_i all have expected value \$240, $E[X] = 10000(\$240) = \$2,400,000$. By the assumption that the X_i are independent, we have $\text{Var}(X) = \text{Var}(X_1) + \cdots + \text{Var}(X_{10000})$. Since the X_i all have variance 800^2 , $\text{Var}(X) = 10000(800^2) = 6,400,000,000$ and $\sigma(X) = 100(\$800) = \$80,000$.

Using the normal approximation, we have $P(X \geq 2,700,000) \approx P(\text{standard normal} \geq 3.75)$. The z-score 3.75 is not in the normal table in Durrett, though it is in the table that I handed out in class. Using that table, we get an estimate of 0.0001 for the probability. We can also use the approximation method from Example 6.28 to estimate $P(\text{standard normal} \geq 3.75)$. The formula from that example gives an upper

bound on that probability of $\frac{1}{\sqrt{2\pi}} \frac{1}{3.75} e^{-\frac{3.75^2}{2}} \approx 0.000094$. Either way, this probability is quite small!

The small probability of total claims at least \$2.7 million illustrates the core principle of insurance: if the claims are independent, then the company can predict the total annual claim fairly precisely, even though individual claims may vary a lot. The danger of this principle is in the independence assumption. If the company does not balance its policies well, independence may fail dramatically. Imagine, for example, an insurance company that insures all the homes in a place that has a devastating forest fire.

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Let T_i be the time it takes to serve customer i . Let $X = T_1 + \cdots + T_{49}$ be the total time it takes to serve 49 customers. We want to estimate $P(X \leq 240)$ (measuring time in minutes).

We assume that the T_i are iid with mean 5 minutes—too long! Has Durrett ever been to a grocery store?—and standard deviation 2 minutes. We then have $E[X] = 49(5) = 245$ and $\text{Var}(X) = 49(2) = 98$. Since X is the sum of a large number of iid random variables, it is reasonable to expect it approximately follows a normal distribution.

$$\begin{aligned} P(X \leq 240) &\approx P(\text{standard normal} \leq -0.505) = P(\text{standard normal} \geq 0.505) \\ &= 1 - P(\text{standard normal} \leq 0.505) \approx 1 - 0.6950 = 0.3050. \end{aligned}$$

The exercises in the next group are all about testing probability models. In each exercise, Durrett implicitly describes a probability model for the process that generates some data. To solve the exercise, you will calculate a suitable p-value, the probability that Durrett's model would generate data like the given data or more extreme.

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In this question we are testing the probability model that the dice rolls are independent and that for each roll, the values 1 through 6 are equally likely.

With this model, the number N of 6s in 18000 rolls has the distribution $N \sim \text{binomial}\left(18000, \frac{1}{6}\right)$.

We thus have $E[N] = 18000 \left(\frac{1}{6}\right) = 3000$, $\text{Var}(N) = 18000 \left(\frac{1}{6}\right) \left(\frac{5}{6}\right) = 2500$, and $\sigma(N) = 50$. Since N is the sum of 18000 iid indicator variables (indicating “6 on roll i ”), it is reasonable to treat N as approximately normal.

Because the observed number of 6s, 3123, is larger than the expected value, we calculate a right-tail p-value, $P(N \geq 3123)$:

$$\begin{aligned} P(N \geq 3123) &= P(N \geq 3122.5) \approx P(\text{standard normal} \geq 2.45) = 1 - P(\text{standard normal} \leq 2.45) \\ &\approx 1 - 0.9929 = 0.0071. \end{aligned}$$

This p-value is very small, which means that data from the probability model are quite unlikely to look like the data generated by the casino employees (3123 or more 6s in 18000 rolls). The casino owner would thus reject the “fair die” model.

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We test the “not cheating” model to see if it is plausible that this model could yield 9 honor cards (last card dealt) in 16 deals. In this model, for any deal, the probability that an honor card is at the bottom of the deck is $16/52$, since there are 16 honor cards (four of each suit), and all 52 cards are equally likely to end up at the bottom after shuffling. We assume that the events “honor card at bottom in deal i ” are independent.

With this probability model, the number N of honor cards in 16 deals has $N \sim \text{binomial}\left(16, \frac{16}{52}\right)$. We thus have $E[N] = 16 \left(\frac{16}{52}\right) = \frac{256}{52} \approx 4.92$, $\text{Var}(N) = 16 \left(\frac{16}{52}\right) \left(\frac{36}{52}\right) \approx 3.41$, and $\sigma(N) \approx 1.85$. The distribution of N will be roughly normal. Because the observed number of honor cards is larger than the expected value, we calculate a right-tail p-value, $P(N \geq 9)$:

$$\begin{aligned} P(N \geq 9) &= P(N \geq 8.5) \approx P(\text{standard normal} \geq 1.94) = 1 - P(\text{standard normal} \leq 1.94) \approx 1 - 0.9738 \\ &= 0.0262. \end{aligned}$$

Since this p-value is small (less than the standard cutoff of 0.05 from statistics), we conclude that the “not cheating” probability model does not often produce data like what we observed. We shouldn't conclude that the player is cheating, since we have just cast doubt on the probability model. There could be other reasons that this probability model does not match the data. (Maybe we are not shuffling as well as we should, for example.)

Mendelian genetics is a probability model for inheritance. It suggests that offspring of two genotype Aa parents have a probability 1/4 of having genotype aa. According to this model, the number N of aa genotype offspring out of 96 offspring of Aa parents has the binomial(96, 1/4) distribution. Therefore, $E[N] = 96(1/4) = 24$, $\text{Var}(N) = 96(1/4)(3/4) = 18$, and $\sigma(N) \approx 4.24$.

The random variable N is a sum of a large number (96) of iid indicator random variables (indicating the event “offspring i has aa genotype”). Since the observed number of aa offspring (30) is larger than the expected number (24), we calculate a right-tail probability as the p-value:

$$\begin{aligned} P(N \geq 30) &= P(N \geq 29.5) \approx P(\text{standard normal} \geq 1.30) = 1 - P(\text{standard normal} \leq 1.30) \\ &\approx 1 - 0.9032 = 0.0968. \end{aligned}$$

Since this p-value is larger than the conventional statistics cutoff of 0.05, we conclude that the data are compatible with the Mendelian inheritance model.

We interpret the softball player’s brag (being a 0.300 hitter) to mean that in any at-bat he (she?) hits with probability 0.300. We assume, perhaps unreasonably, that the events “hit on at-bat i” are independent. Under this assumption, we can think of the number N of hits in 84 at-bats as a random variable with the binomial(84, 0.300) distribution. We thus have $E[N] = 84(0.300) = 25.2$, $\text{Var}(N) = 84(0.300)(0.700) = 17.64$, and $\sigma(N) = 4.2$.

The random variable N is a sum of 84 iid indicator random variables (indicating the events “hit on at-bat i”), and so it is reasonable to treat N as approximately normally distributed. Since the number of hits (21) was less than expected (25.2), we calculate a left-tail probability for the p-value testing the player’s model.

$$\begin{aligned} P(N \leq 21) &= P(N \leq 21.5) \approx P(\text{standard normal} \leq -0.881) = P(\text{standard normal} \geq 0.881) \\ &= 1 - P(\text{standard normal} \leq 0.881) \approx 1 - 0.8106 = 0.1894. \end{aligned}$$

Since the p-value is larger than the usual cutoff (0.05) from statistics, we conclude that the data are compatible with the player’s probability model. It is plausible that the player is a 0.300 who had a string of bad luck.

The probability model proposed in the question is that for each day in Ithaca (New York), the probability of rain is 1/3 and that the events “rain on day i” are independent. Under this model, the number N of rainy days in a 60-day period is a random variable with the binomial(60, 1/3) distribution. Therefore, $E[N] = 60(1/3) = 20$, $\text{Var}(N) = 60(1/3)(2/3) = 40/3$, and $\sigma(N) \approx 3.651$.

Since N is the sum of a large number (60) of iid random variables (indicators for “rain in Ithaca on day i”), it is reasonable to expect that N is approximately normally distributed. The number of rainy days (13) was less than expected, and so to test the probability model, we calculate a left-tail p-value, $P(N \leq 13)$:

$$\begin{aligned}P(N \leq 13) &= P(N \leq 13.5) \approx P(\text{standard normal} \leq -1.78) = P(\text{standard normal} \geq 1.78) \\&= 1 - P(\text{standard normal} \leq 1.78) \approx 1 - 0.9625 = 0.0375.\end{aligned}$$

Since the p-value is smaller than the conventional cutoff of 0.05, we judge the data to be incompatible with the probability model.